

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India 09/2021 – 01/2025 (3.5 Years)

- Defined and implemented an **AI strategy** for a Bitcoin mining operations platform, translating operational challenges into data and optimization problems solvable by **ML models**, resulting in a projected **15% increase in site-level efficiency**.
- Architected and delivered a **standardized data baseline** for mining-site telemetry and events, ensuring data was structured, normalized, and accessible for **AI/ML use cases**, improving data ingestion quality by **20%**.
- Developed and deployed **predictive maintenance models** using **Python** and **scikit-learn**, integrating them into operational workflows to forecast hardware failures, reducing unscheduled downtime by **12%**.
- Engineered **performance optimization models** for Bitcoin mining operations, analyzing energy-efficiency insights and recommending automation strategies that enhanced resource utilization by **10%**.
- Collaborated with backend engineers to evolve the data pipeline and integration standards for the **Mining Operating System (MOS)**, ensuring **AI models** were efficient, reliable, and explainable.
- Established **best-practice frameworks** for scalable, production-ready **AI systems**, documenting architecture and modelling assumptions to support cross-team collaboration and knowledge transfer.

SKILLS

AI/ML & Optimization – Machine Learning, Mathematical Optimization, Predictive Modelling, Anomaly Detection, Performance Optimization, Data Architecture, Data Modelling, AI Strategy, Operational Efficiency

Programming & Scripting – Python, Java, SQL, R

Data Engineering & BI – Data Pipelines, ETL, Data Warehousing, Power BI, DAX, Power Query, Tableau, Database Schema Design

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, CI/CD, Linux Shell Scripting

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn, REST APIs

Methodologies & Tools – Agile, Scrum, Jira, Confluence, Splunk, Dynatrace

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin 02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India 01/2021

PROJECTS

Real-Time Anomaly Detection System for Financial Transactions

- Developed a **Python**-based anomaly detection system to monitor transaction flows on a financial-services dashboard, identifying suspicious patterns in real-time.
- Implemented data pipelines using **pandas** and **SQL** to process high-volume transaction data, ensuring data quality and accessibility for analysis.

- Integrated the system with operational alerting mechanisms, reducing manual review of fraudulent transactions by **30%** and improving incident response times.

Predictive Analytics Dashboard for Industrial Equipment

- Built an interactive dashboard using **Power BI** to visualize predictive maintenance insights for industrial machinery, leveraging **DAX** for complex calculations.
- Applied **machine learning fundamentals** from MSc coursework to model equipment failure probabilities based on sensor telemetry and operational logs.
- Designed the data model and ETL processes using **Power Query**, enabling stakeholders to track equipment health and optimize maintenance schedules, contributing to a projected **15% reduction in repair costs**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)